Day 1 — The Sūtra and the Hook

Module: Nikhilam Subtraction · **Band:** Middle · **Time:** 45 min

Learning objective

By the end of this lesson, students will record a personal baseline time for `10 000 − 4567` using standard borrowing, will recite the Nikhilam sūtra in Sanskrit, and will have watched the method solve the same problem visibly faster.

Materials

- Printed baseline sheet: ONE subtraction problem (`10 000 − 4567 = ?`). One per student.
- Stopwatch or wall clock (one per pair is plenty).
- Whiteboard / large paper for the teacher demo.
- A printed Nikhilam sūtra poster — IAST + English — on the wall.

Vocabulary introduced today

- *Nikhilaṁ Navataścaramaṁ Daśataḥ* (IAST: nikhilaṁ navataścaramaṁ daśataḥ; lit. “all from nine and the last from ten”) — the sūtra we’ll learn this fortnight.
- *sūtra* (IAST: sūtra; lit. “thread, formula, aphorism”) — a short mnemonic rule. Tirthaji’s *Vedic Mathematics* gives 16 of them.

Lesson flow

Warm-up (5 min)

- Teacher: *“How long does it take you to solve* `10 000 − 4567`*? Make a guess.”*
- Take 3–4 guesses. Write the range on the board (e.g. “15 sec to 60 sec”).
- Quick make-10 reflex round: teacher calls digits; class chorus the partner. 8 → 2, 6 → 4, 9 → 1, 3 → 7, 5 → 5, 4 → 6, 7 → 3, 2 → 8.

Baseline timing (8 min)

- Hand out the baseline sheet face-down.
- Students write their name and predicted time on the back.
- On “Go,” flip and solve using whatever method they normally use (standard borrowing).
- Record their actual time (the wall clock is running).
- Mark answer; honest self-correction — they’re racing themselves over the module, not each other.

Meet the sūtra (5 min)

- Point to the wall poster. Teacher reads slowly:

  Nikhilaṁ Navataścaramaṁ Daśataḥ “All from nine and the last from ten.”

- Class echoes 3 times: soft / normal / loud.
- Teacher: *"That's the whole rule. Six Sanskrit words. By Day 10 you'll use it without thinking."*

Teacher demonstration (12 min) — the same baseline problem, the Nikhilam way

Write on the board:

```
   1 0 0 0 0
-        4 5 6 7
```

Narrate:

1. *"We want to subtract 4567 from 10 000. That's 10^4 — a clean power of ten. Perfect Nikhilam territory."*
2. *"Treat 4567 as a 4-digit number. Apply the rule digit-by-digit."*

Step	Apply rule	Get
First digit (4):	"from 9"	9 − 4 = **5**
Second digit (5):	"from 9"	9 − 5 = **4**
Third digit (6):	"from 9"	9 − 6 = **3**
Last digit (7):	"from 10"	10 − 7 = **3**

3. Read the answer top-to-bottom: **5433**.
4. **Verify** (with the class): 4567 + 5433 = 10 000. ✓
5. *"Notice — I never borrowed. Not once. The 'all from nine' rule means we're always subtracting from 9, which is bigger than any single digit."*

Activity (10 min) — First try

- Students re-do the same problem (`10 000 − 4567`) on the back of their baseline sheet, using the Nikhilam rule. Refer to the wall poster.
- Time themselves again.
- Compare: how much slower (or faster) was the Nikhilam method on the FIRST try?

Wrap-up (5 min)

- Show of hands: who was faster with Nikhilam on first try? Who was slower? (Most will be slower — that's expected. Practice fixes it.)
- Preview Day 2: *"Tomorrow we drill the foundational skill — finding the complement of a 2-digit number to 100, fast."*
- Set the wall-poster Sanskrit chant as the daily opener for the next 9 days.

Student-facing content

The Nikhilam Sūtra

Nikhilaṁ Navataścaramaṁ Daśataḥ — “all from nine and the last from ten.”

To subtract from 10, 100, 1000, 10 000, …: 1. Subtract each digit *from 9*, **except the last digit** — subtract it *from 10*. 2. Read the answer top-to-bottom. 3. Verify: original + answer = the round base.

No borrowing. Ever.

Today’s baseline: - My time with my normal method = ______ seconds - My time with Nikhilam (first try) = ______ seconds

I expect the second number to drop by Day 5.

Homework

- Memorise the sūtra. Be able to say it from memory tomorrow.
- Practice: find the complement of these 2-digit numbers to 100:
  - 73, 41, 28, 86, 95, 17, 60, 52, 34, 89.
  - Time yourself. Aim for under 30 seconds for all ten.

Differentiation

- **Tier up:** Try Nikhilam on `100 000 − 23 456`. Time it.
- **Tier down:** If the 4-digit problem feels overwhelming, drill complements to 100 (2-digit numbers) for the activity instead.

Teacher notes

- The baseline time is the *single most important* artefact of Day 1. Make sure every student records it honestly. Day 5’s retest depends on it.
- Don’t promise a specific speedup — say “you’ll likely be faster by Day 5; how much depends on you.”
- The Sanskrit pronunciation is hard. Model it slowly; let students laugh on the first attempt. By Day 5 they’ll all say it cleanly.
- One paraphrase from Tirthaji that helps: “Nikhilam (निखिलं) means ‘whole,’ ‘entire,’ ‘all.’ Navataḥ means ‘from nine.’ Caramaṁ means ‘the last.’ Daśataḥ means ‘from ten.’ Six words; one method.”

Citation(s) used in this lesson

- Tirthaji, *Vedic Mathematics* (1965), sūtra Nikhilam (#2). The full sūtra reads *Nikhilaṁ Navataścaramaṁ Daśataḥ*.
- *Microsoft Word - sutras1.pdf* — wall-poster source.